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LOW WATER EXCHANGE SHRIMP FARMING: IMPROVEMENTS IN THAILAND

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How the implementation of heterotrophic, low water exchange shrimp-farming systems – including the use of biofloc – in Thailand has helped to limit the spread of pathogens and improved farm performance.

Between 1970 and 2015, the shrimp industry in Thailand experienced several cycles of rapid expansion and intensification of production, followed by sector-wide losses from a series of viral disease outbreaks. Farms were forced to innovate and adapt management measures to address the perceived sources, or drivers, of disease. These measures were either developed empirically or scientifically, based on field observations, research and epidemiological results. This resulted in changes in farm location, water supply and exchange, aeration and water circulation, sediment management, feeding, plankton bloom control, stocking and harvest schedules, species cultured and biosecurity measures.

Large-scale water exchange, typically up to 20 percent per pond per day, started to decline in Thai shrimp culture in the early 1990s, as a response to successive waves of viral disease outbreaks – including yellow head virus (YHV) and white spot syndrome virus (WSSV) disease – with infections and farm-to-farm transmission attributed to water intake from shared water systems.

By the late 1990s, closed shrimp culture systems were in place throughout Thailand, usually with a single pond, because most farms were small scale. Larger farmers with multiple ponds excavated reservoirs and treated incoming water. It was only realized later that disease pathogens were entering via the incoming shrimp postlarvae (PLs), which led to the screening and disease testing of incoming PLs, although in reality prevention of pathogen entry was practically impossible, at least until the development of specific pathogen free (SPF) shrimp which came much later.





Over time Thai shrimp farmers increased their understanding of how to use the directional placement of paddle wheel aerators to improve water circulation within ponds and were able to operate shrimp production units with only minimal water exchange. Feeding frequency and diets were adjusted to give more stable feeding regimes and reduced dissolved oxygen fluctuation in grow-out ponds. Culture cycles were also shortened to avoid going beyond the carrying capacity of the ponds, as had previously happened when some Thai shrimp farmers pushed their systems too far.

Subsequent shrimp disease outbreaks entrenched the closed low water exchange shrimp culture system in Thailand. Extremely high levels of water circulation within the culture pond increased system stability, but the downside was significant bank erosion due to water movement. This led to the use of pond liners, which reduced pond erosion and stopped disease pathogen build up, as sediment was previously a 'sink' that housed pathogens.

Thai farmers then developed central drains and cleaned their ponds by using aerators to create circular flows to carry wastes towards these drains. With limited water exchange, they topped up pond water from closed on-farm reservoirs.

The heterotrophic pond system emerged in Thailand in the 1990s and mainly took place on medium-sized shrimp farms with sufficient pond numbers, some of which were used as water-holding and treatment reservoirs. With the introduction of SPF whiteleg shrimp (*Penaeus vannamei*) in the mid-2000s there was a further reduction of viral pathogens and a period of sustained shrimp productivity in Thailand, until the outbreaks of acute hepatopancreatic necrosis disease (AHPND) and the fungal microsporidian parasite *Enterocytozoon hepatopenaei* (EHP).

In 2010, the average productivity of intensive Thai shrimp farms was 10.6 tonnes of shrimp per hectare per year. In 2011, Thailand produced 600 000 tonnes of shrimp and was the world's leading shrimp exporter. With the outbreak of AHPND in Thailand that year, caused by *Vibrio parahaemolyticus* bearing two toxin genes located at the 69 kbp plasmid, Thai shrimp production dropped 63 percent, falling to 220 000 tonnes in 2015. To overcome AHPND Thai shrimp farmers closed their farms and ran them as closed recirculation systems, with one-third of ponds used as reservoir volume, one-third for shrimp production and one-third as natural biological treatment ponds.

A second disease issue, the spread of EHP in both whiteleg shrimp and giant tiger prawn (*Penaeus monodon*) further encouraged Thai shrimp farmers to minimize water intake from outside the farm, and to operate grow-out farms by recycling and biologically treating water.



TECHNIQUE AND APPROACH USED

Thai shrimp farmers have changed several management practices to overcome AHPND and EHP. Techniques include:



on-farm recirculation and natural treatment of pond water



a move to lined ponds



on-farm nursing of PLs in knock-down nursery tanks to provide larger PLs for stocking into grow-out ponds



use of lower protein diets, combined with the addition of a carbon source to promote the proliferation of heterotrophic rather than autotrophic bacteria



use of strong aeration and water currents to keep biofloc in suspension

In-farm recycling means that there is no water intake for significant periods of time, which keeps water salinity within ideal ranges for the culture species. Another benefit of reduced water exchange on shrimp and fish grow-out farms is that it is much easier and cheaper to remove the nutrients excreted by the cultured animals, as the volume of discharged water is much lower.



Larval shrimp nursery biofloc



Microbiofloc in an Imhoff cone

The move from giant tiger prawn to whiteleg shrimp also facilitated the adoption of heterotrophic biofloc systems, because whiteleg shrimp are effective detrital feeders. The provision of appropriate structures, like vertically hanging mats, increases shrimp survival and yield, and reduces food conversion ratios (FCRs) in biofloc production systems (McNeil, 2000).

SCOPE AND SCALE OF APPLICATION

The heterotrophic biofloc system is now widely used, but has been taken to another level with tailored feeds with reduced protein content, because the natural biofloc on which the shrimp feed is protein-rich. The reduced pond sediment erosion of lined ponds also makes the floc more easily accessible for the shrimp to graze on.



HOW THE HETEROTROPHIC POND SYSTEM MANAGEMENT WORKS

As shrimp production cycles progress, increased sediment loads and nutrients leached from feeds result in deterioration of water quality that promotes the proliferation of pathogens, including the pathogenic *Vibrio* species. Promoting phytoplankton can improve water quality, but is hard to manage – with self-shading and a reduction in sunlight penetration often resulting in phytoplankton die-offs, unless intense aeration is provided. When die-offs occur the sudden fluctuations in dissolved oxygen and pH stress the cultured animals. Biofloc systems get around this problem by using micro-organisms to efficiently remove and recycle nitrogen compounds and nutrients from organic and inorganic matter (unconsumed and non-digested feed, metabolic residues and carbon sources) into new microbial cells. The floc makes ponds equivalent to activated sludge ponds or bioreactors.

A biofloc system requires sunlight, a carbohydrate source and vigorous aeration. Through photosynthesis, primary-producing autotrophic and heterotrophic bacteria convert uneaten feeds, faeces and nutrients leached from feeds and break down toxic ammonia and nitrates, while rapidly replicating and attracting a growing host of organisms, including diatoms, fungi, algae, protozoans and various types of plankton. Loosely bound by bacterial mucous, most of the floating clumps or ‘flocs’ are microscopic. Larger aggregations resembling brown or green sludge can be seen by the human eye. The flocs are a good food source for both shrimp and fish.

The micro-organisms produce *in situ* microbial protein, which is rich in natural protein-lipids on which the shrimp can feed 24 hours per day. A cheap carbon source like molasses is provided to keep carbon-to-nitrogen ratios high (15–20), so that heterotrophic micro-organisms predominate over autotrophs in the pond water. Vigorous oxygenation is provided to keep the biofloc in suspension and to keep dissolved oxygen levels above 4 mg/L (Emerenciano, Gaxiola and Cuzon, 2013). High dissolved oxygen concentrations are required both for the shrimp and to support aerobic decomposition of organic matter and nitrification by bacteria (Boyd and Clay, 2002).

The ponds start with a phytoplankton bloom (green water), but after 8 to 10 weeks the waters become turbid, with particles covered by bacteria. Turbidity caused by the floc reduces light penetration and prevents the growth of phytoplankton after the first few weeks of culture, and the water colour changes from green to brown – then even black – as the bacterial load increases many times. Nitrogen in the feed increases total ammonia nitrogen concentrations, but nitrifying bacteria rapidly convert ammonia nitrogen to nitrate, which results in steady and low total ammonia concentrations in the region of 1 to 2 mg/L, until harvest. Limestone is usually applied to maintain the alkalinity of the ponds at 80 mg/L or higher, and this keeps the water pH stable at 7.2 to 7.4 (Boyd and Clay, 2002).



COMPLEXITY AND ACCESSIBILITY

Heterotrophic biofloc systems have been adopted by the intensive shrimp culture sector and the intensive tilapia culture sectors in Thailand in particular, but also in many other Asian countries.

The main reasons are that:

- Biofloc systems are extremely stable, with consistent water quality conditions, which reduces the likelihood of pathogenic disease outbreaks.
- The addition of a carbon source allows the protein content of the pelleted feeds to be reduced, because the protein content of the biofloc is high. In addition, shrimp reared in biofloc retain more of the nitrogen from the feed.
- Biofloc systems are more robustly profitable.
- The natural probiotic properties of biofloc give aquatic animals greater resistance to several disease pathogens, including *Vibrio* spp.

There has been strong promotion of heterotrophic biofloc systems for shrimp in particular – but also for other aquatic species, including tilapia – through informal networks like provincial shrimp clubs and also commercial networks hoping to profit from the dissemination of knowledge on new and effective management techniques.

For biofloc systems, it is imperative to have a reliable and constant source of electricity, to provide strong aeration and water current movement to keep the biofloc in suspension at all times. As a result, back-up generators are required in case of power cuts. This means that biofloc technology is better suited to medium- and large-scale intensive farms.

THE OUTCOME AND BENEFIT

Disinfection of aquaculture production units like tanks and ponds creates a near-sterile environment which provides conditions that initially support the dominance of opportunistic r-strategy bacteria, including dangerous *Vibrio* species. If *Vibrio* levels are allowed to increase unchecked, pathogen virulence of *Vibrio* species can be triggered by quorum sensing.

Biofloc systems improve stock health by improving water quality – with the continuous aeration providing stable dissolved oxygen and pH levels – creating an increase in the amount of bio-available natural food organisms, strengthening the immune systems of fish and shrimp, and reducing disease losses.

In trials which held whiteleg shrimp in either clear water, filtered biofloc water or in biofloc water for 5 to 10 days prior to challenging them with *Vibrio parahaemolyticus* (VPAHPND) the highest survival rate was recorded for shrimp reared and challenged in unfiltered biofloc. The results demonstrate that biofloc can protect whiteleg shrimp from VPAHPND and that the management of biofloc in aquaculture ponds can assist in controlling bacterial infections (Shinn *et al.*, 2018).

Following feedback from Thai farmers that farms operated with biofloc or lower salinity culture water with Nile tilapia (*Oreochromis niloticus*) in their treatment reservoirs had less AHPND infection, Sajali *et al.* (2019) conducted a trial to test if biofloc- and Nile tilapia-conditioned water impacted the survival of whiteleg shrimp when challenged with VPAHPND. The results indicated that this provided some protection against AHPND and suggest that the implementation of inexpensive pond management strategies in the shrimp industry may assist in reducing the impact of VPAHPND. The tilapia-conditioned water component of this study merits further applied trials (A. Shinn, personal communication, 2021). Having stable microbial ecosystems is believed to make integrated systems producing the likes of shrimp, tilapia and seaweed more resilient to pathogenic disease outbreaks.

The addition of a carbon source like molasses and the use of vigorous aeration and stronger water currents within ponds promote heterotrophic organisms and biofloc production, and encourage the domination of generally harmless and preferred k-strategist bacteria over r-strategist bacteria in culture units, which reduces the risk of disease outbreaks.

Biofloc systems recycle protein and convert ammonia into microbial proteins that are consumed by filter feeders and increase the proportion of protein content in feeds that is utilised from about 25 percent to 50 percent, thereby saving farmers money. Shrimp reared in biofloc systems also retain more of the nitrogen from the commercial feed (Boyd and Clay, 2002).

Research in Bangladesh has shown that lowering the digestible protein to digestible energy ratio in pond feeds, by incorporating less nutrient-dense crop wastes in the fish feeds, improved the contribution of natural food to fish production in ponds, as well as the benefit to cost ratio by 22 percent (Kabir *et al.*, 2019; Kabir *et al.*, 2020).

Detritivorous shrimp, tilapia and common carp consume plant and animal remains and heterotrophic bacteria. Reduced particle size allows bacteria and fungi to break the materials into their constituent chemical parts for recycling. Lower trophic species like detritivores require less energy to grow, as typically 90 percent of the available energy is lost each time it passes through a trophic level.

In early 2000, Belize Aquaculture Limited achieved an average shrimp production of 11 231 kg/ha/crop, with the average crop cycle taking 139 days for whiteleg shrimp using a super-intensive biofloc system and zero water exchange. The highest productivity achieved was 27 200 kg/ha from 650 m² ponds aerated with 60 hp/ha and containing 1 350 m² of vertical surfaces. The increased surface area improved shrimp survival and reduced FCRs as well as the feed required by 30 percent (Boyd and Clay, 2002).

In summary, heterotrophic biofloc low water exchange production systems can produce more aquatic foods sustainably, with minimal water exchange, less sludge and pond wastes and also improve farm profitability. Biofloc systems provide both a low-cost and natural method of treating and improving the water quality in grow-out ponds, and simultaneously provide an additional cost-saving food source.



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